

Phaco fluidics, ultrasound and OVD behaviour

Part 2 | Flow and vacuum are different control variables

Know what the machine is controlling before interpreting occlusion.

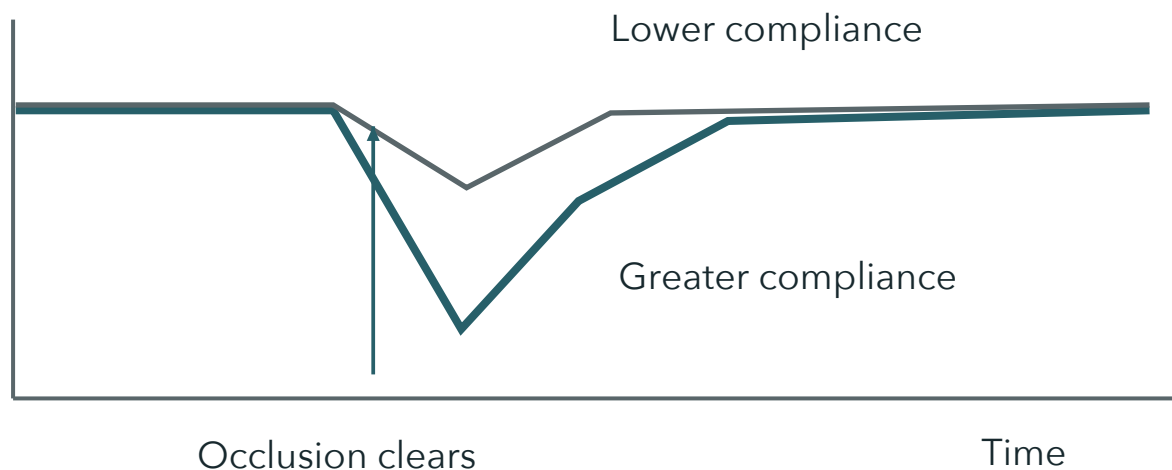
Pump behaviour	Interpretation
Peristaltic roller-driven flow	Vacuum rises substantially with tip occlusion
Venturi vacuum generation	Flow follows the pressure gradient
Infusion pressure	Supports inflow, not the same as aspiration
Wound leakage	Changes chamber balance without changing pump label

Settings interact with tip geometry, tubing and the eye. Neither pump name guarantees a safe chamber in every state. During occlusion and release, ask whether outflow is temporarily exceeding available inflow; ultrasound power is a separate control and cannot correct a fluidic imbalance.

Occlusion stores a pressure-volume disturbance

Compliance determines how much fluid is drawn when the blockage clears.

Chamber pressure



Original qualitative traces: greater compliance can produce a deeper transient chamber-pressure fall.

Laboratory comparisons show why aspiration-circuit compliance and capacity matter. During occlusion, deformable components store volume change; release can rapidly draw fluid from the chamber. Low compliance is advantageous under otherwise comparable conditions, but bench findings are not a guarantee of identical clinical behaviour across every setting or eye.

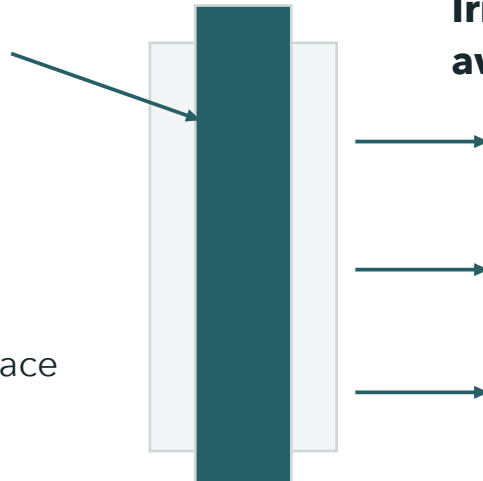
Cooling requires actual fluid movement

Energy delivery, sleeve fit and occlusion can interact.

Vibrating needle

Irrigation carries heat away

Incision / sleeve interface



Restricted flow reduces cooling.

Original section: irrigation removes heat around a vibrating needle at the incision.

A tightly compressed sleeve or obstructed flow can reduce heat removal and cause incisional whitening, contraction and wound damage during ultrasound delivery. Reassess flow and mechanical restriction rather than continuing power through an unsafe wound. Thermal injury may leave leakage and substantial astigmatism even after the nucleus is removed.

OVD behaviour explains both benefit and removal burden

Space maintenance and endothelial coating are related but distinct goals.

Behaviour	Typical tradeoff
Cohesive	Maintains space; tends to leave as a bolus
Dispersive	Coats tissue; can persist during aspiration
Combined strategy	Balances space with endothelial protection
Residual OVD	Can contribute to postoperative pressure elevation

Choose and remove OVD with its rheology in mind. Persistence along the endothelium can be beneficial during flow but makes complete removal less straightforward. The material that maintains a deep chamber most readily is not automatically the one with best sustained surface coating.

Iris prolapse is a pressure and tissue problem

Repeatedly pushing iris back without changing the cause can fail.

- Pause and identify chamber pressure, wound and iris behaviour.
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- Reduce unsafe pressure gradients and assess posterior causes.
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- Reposition tissue gently using an appropriate specialist technique.
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- Address wound geometry or pupil instability before resuming.

Consider incision configuration, small pupil or IFIS, and positive posterior pressure. An iris that repeatedly re-enters a wound is signaling an unresolved mechanism. Mechanical expansion and adjusted fluidics can help selected cases; avoid forceful tissue manipulation that trades temporary access for iris damage.